10.12 Molecular Orbital Theory

In 1932, Hund and Mulliken had put forward the molecular orbital theory. The main features of this theory are as follows

(i) New orbitals termed as the molecular orbitals are formed by the overlap of atomic orbitals of the combining atoms. The atomic orbitals lose their individual identity after the formation of the molecular orbitals.

(ii) The number of molecular orbitals formed is equal to the number of atomic orbitals involved in their formation.

(iii) The electron in an atomic orbital is influenced by just one positive nucleus of the atom, that is, it is monocentric, whereas depending upon the number of atoms in the molecule the electron of a molecular orbital is under the influence of more than one nuclei, that is, it is polycentric.

(iv) Similar to the atomic orbitals, the filling of electrons in the molecular orbitals also follows the Aufbau principle, Hunds rule of maximum multiplicity and Pauli's exclusion principle.

(v) Atomic orbitals with comparable energies as well as proper orientations combine to form molecular orbitals.

Linear Combination of Atomic Orbitals (LCAO)

According to wave mechanics, the atomic orbitals can be expressed by wave function (ψ), which represent the amplitude of the electron waves. The values of these wave functions can be obtained by the solutions of Schrodinger wave equation. Just like atoms, the Schrodinger wave equation can also be applied to molecules. The wave function for molecular orbitals is obtained by the linear combination of atomic orbitals (LCAO).

To explain the LCAO concept, let us consider two atoms A and B having the wave functions ψ_A and ψ_B. According to the LCAO concept, molecular orbitals are formed by the linear combination of atomic orbitals. The combination can take place by addition (constructive interaction) or subtraction (destructive interaction) of the wave functions of atomic orbitals. These two types of combinations give rise to two molecular orbitals as described below.

(i) Bonding molecular orbitals are obtained by the addition of wave functions of atoms and may be represented as

$$\psi\,(\text{MO}) = \psi_A + \psi_B \quad \text{(Constructive interaction).}$$

The bonding MO has energy lower than the energy of the atomic orbitals from which it is formed and this difference of energy is termed as the stabilisation energy.

(ii) Antibonding molecular orbitals are obtained by the subtraction of wave function of atoms and may be represented as

$$\psi^*\,(\text{MO}) = \psi_A - \psi_B \quad \text{(Destructive interaction).}$$

An asterisk * represents the antibonding molecular orbital. The antibonding molecular orbital has energy higher than that of the constituent overlapping atomic orbitals. The difference in energy between the antibonding molecular orbital and the combining atomic orbitals is called the destabilisation energy. Thus, antibonding MO destabilises the molecule. *It is important to note that the bonding MO is stabilised to the same extent as the antibonding MO is destabilised* (Fig. 10.36).

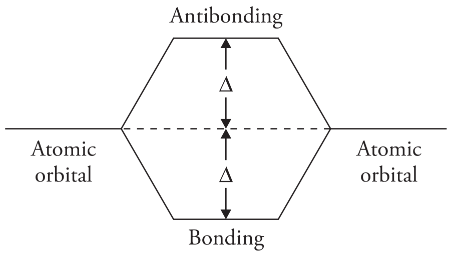


Figure 10.36 *Formation of bonding and antibonding molecular orbitals*

Moreover, when bonding MO is formed the two waves are in phase and the amplitude gets added up. Therefore, the resulting wave function has more electron density in between the nuclei. On the other hand, in the formation of antibonding MO the two waves are out of phase, there is destructive interference and the amplitudes of the waves get subtracted or cancelled. This gives rise to nodal plane between the two nuclei.

Probability Density in Bonding and Antibonding MOs

The probability density in bonding and antibonding molecular orbitals is given by the square of wave function (ψ^2).

For Bonding MO

$\psi = \psi_A + \psi_B$, so that

$$\psi^2 = (\psi_A + \psi_B)^2 = \psi_A^2 + \psi_B^2 + 2\psi_A \psi_B$$

Thus, ψ^2 is greater than $\psi_A^2 + \psi_B^2$ (sum of the probability densities of individual atoms) by an amount equal to $2\psi_A\psi_B$. Therefore, the probability of finding electrons in bonding MO is greater than in either of the atomic orbitals (ψ_A or ψ_B)

For Antibonding MO

$\psi^* = \psi_A - \psi_B$, so that

$$\psi^{*2} = (\psi_A - \psi_B)^2 = \psi_A^2 + \psi_B^2 - 2\psi_A \ \psi_B$$

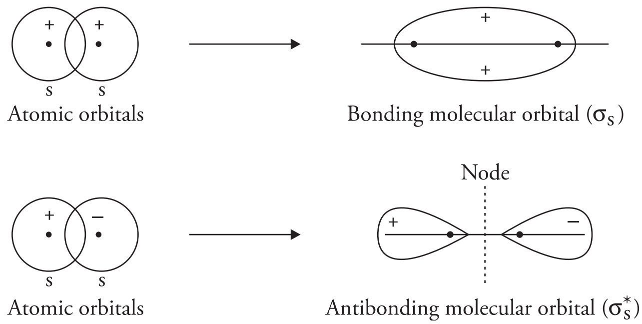
Thus, ψ^{*2} is less than $\psi_A^2 + \psi_B^2$ by an amount equal to $2\psi_A\psi_B$. Therefore, the probability of finding the electrons in antibonding MO is less than in either of the atomic orbitals (ψ_A or ψ_B).

Molecular orbitals from 1s, 2s and 2p atomic orbitals

When two atomic orbitals overlap, along the internuclear axis, then the resulting molecular orbitals are the sigma (σ) molecular orbital. On the other hand, if the atomic orbitals overlap sideways then the pi(π) molecular orbital is formed.

Combination of s-s atomic orbitals

1s with 1s or 2s with 2s



[Note that + and – are signs of wave function and have nothing to do with electrical charges.]

Figure 10.37 *s–s overlap of orbitals*

Combination of s-p atomic orbitals

Combination of s orbital with p orbital occurs provided the lobes of p are directed along the internuclear axis. Bonding MOs are formed when lobes of the same sign overlap and antibonding MO are formed when lobes of opposite signs overlap.

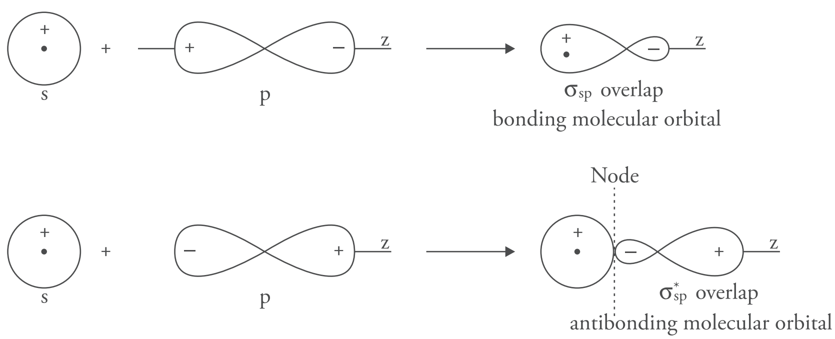


Figure 10.38 *s–p overlap of orbitals*

Combination of p–p atomic orbitals

There are three types of p orbitals directed along the cartesian coordinates *x*, *y*, and *z*, respectively. By convention the *z*-axis is assumed as the internuclear axis. The $2p_z$ atomic orbitals overlap along the internuclear axis forming σ and σ^*molecular orbitals whereas $2p_x$ and $2p_y$ orbitals overlap collaterally forming π and π^* molecular orbitals.

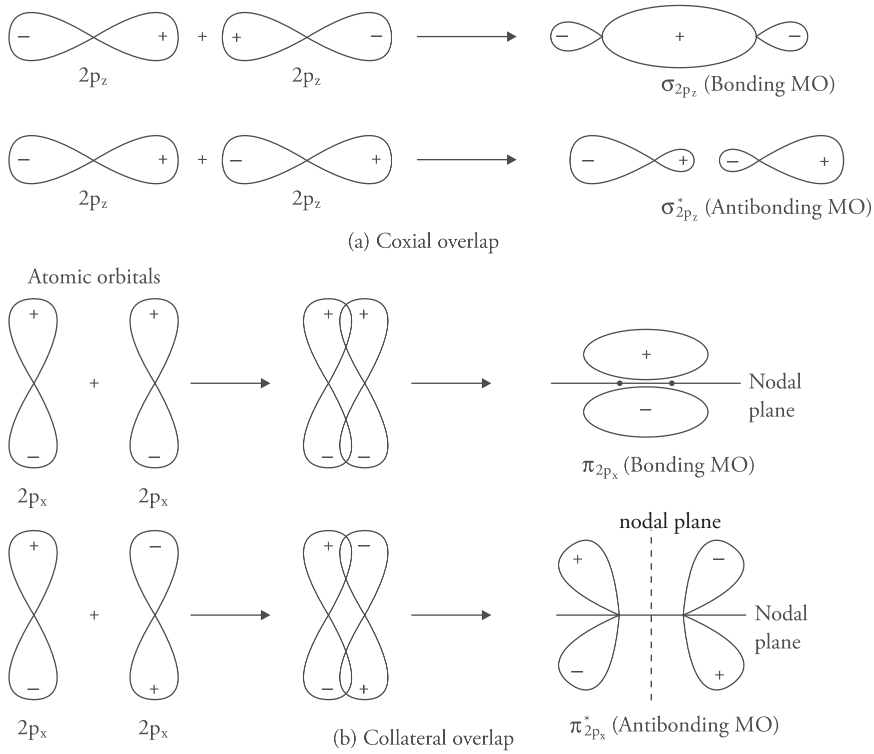


Figure 10.39 *p–p overlap of orbitals: (a) coaxial overlap; (b) collateral overlap*

Combination of p–d orbitals

Bonding and antibonding orbitals may also be formed by the overlap of p orbital of one atom with d orbital of the other atom. As the overlapping is lateral, the molecular orbitals are designated as π orbitals.

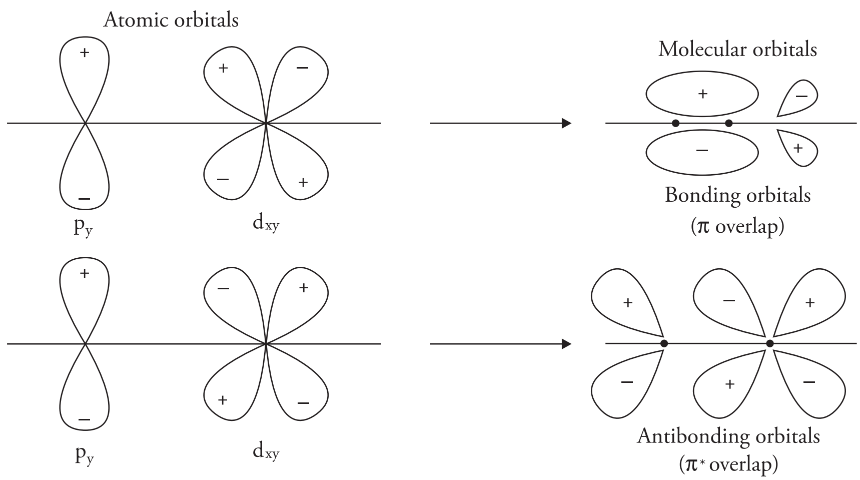


Figure 10.40 *p–d overlap of orbitals*

Combination of d–d orbitals

Bonding and antibonding orbitals formed by the overlap of two d orbitals are termed as δ and δ^*, respectively.

Conditions for the effective overlap of Atomic Orbitals

For effective combination,

(i) The energies of combining atomic orbitals should be nearly same/comparable.

(ii) The extent of overlapping should be large.

(iii) The symmetry of the combining atomic orbitals around the internuclear axis should be the same. For example, 2s atomic orbital of one atom combines with 2s or $2p_z$ orbital of the other atom but not with the $2p_y$ or $2p_x$ orbitals, because when 2s of one atom overlaps with $2p_x$ or $2p_y$ orbital, a collateral overlap occurs and the ++ overlap is cancelled by the + – overlap. Consequently, molecular orbital will not be formed and this situation is termed as non-bonding.

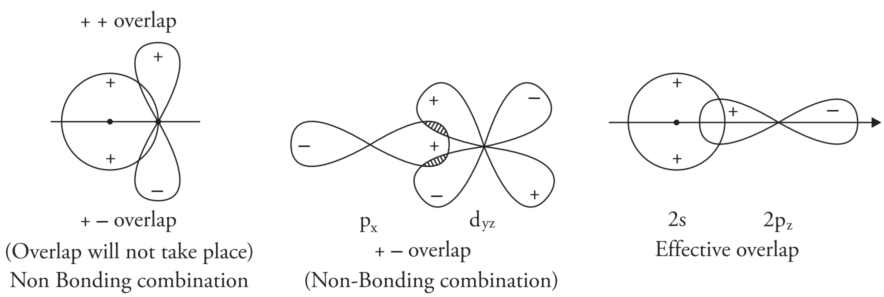


Figure 10.41 *Effective and ineffective overlap of orbitals*

Energy level Diagram for Molecular Orbitals

Spectroscopic methods are used for the experimental determination of the energy of molecular orbitals. The molecular orbitals obtained by the combination of 1s, 2s and 2p atomic orbitals can be arranged in the increasing order of energy as follows

$$\xrightarrow[\text{increasing energy}]{\sigma_{1s},\ \sigma^*_{1s},\ \sigma_{2s},\ \sigma^*_{2s},\ \sigma_{2pz},\ \pi_{2px}=\pi_{2py},\ \pi^*_{2px}=\pi^*_{2py},\ \sigma^*_{2pz}}$$

The energy of bonding π_{2px} and π_{2py} molecular orbitals is exactly similar and they are said to be doubly degenerate. Similarly, π^*_{2px} and π^*_{2py} molecular orbitals also have the same energy and are doubly degenerate.

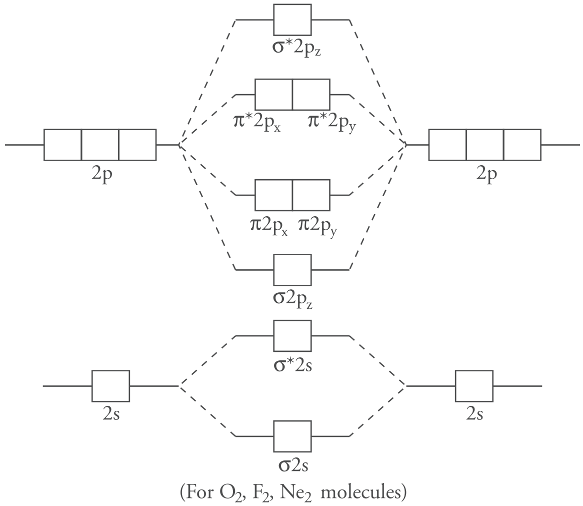


Figure 10.42 *Molecular orbital energy level diagram of O_2, F_2 and Ne_2 molecules*

The molecular orbitals of O_2, F_2 and Ne_2 follow the above-mentioned energy order.

However, as revealed by spectroscopic studies all the molecules do not follow the above-mentioned sequence of energy levels. In Li_2, Be_2, B_2, C_2 and N_2, the energy of the σ_{2pz} molecular orbital is higher than the energy of π_{2px} and π_{2py} molecular orbitals. The molecular orbitals for these homonuclear diatomic molecules have the following order of energy

$$\xrightarrow[\text{increasing energy}]{\sigma_{1s},\ \sigma^*_{1s},\ \sigma_{2s},\ \sigma^*_{2s},\ \pi_{2px}=\pi_{2py},\ \sigma_{2pz},\ \pi^*_{2px}=\pi^*_{2py},\ \sigma^*_{2pz}}$$

The energy level diagram for these molecules is given in Figure 10.43.

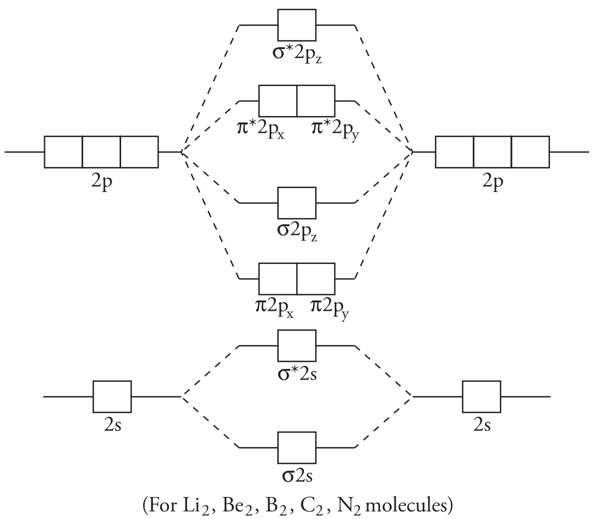


Figure 10.43 *Molecular orbital energy level diagram for Li_2, Be_2, B_2, C_2 and N_2*

Filling of Electrons in Molecular orbitals

The electrons are filled in the molecular orbitals according to the following rules

1. *Aufbau Principle* The electrons enter the molecular orbitals in the order of their increasing energies. Molecular orbitals of lower energy are filled first.
2. *Pauli's Exclusion Principle* This law states that the maximum number of electrons in any molecular orbital is two and the electrons occupying the same orbital must have opposite spins.
3. *Hund's rule* It states that if two molecular orbitals are of the same energy, then one electron goes to each orbital and electron pairing will start only after all the orbitals of equal energy are singly occupied.

Electronic Configuration and Molecular Behavior

The stability of the molecule, its magnetic behavior and strength can be known with the help of molecular orbital diagram.

(i) **Stability of the molecule** Bond formation will occur and the molecule will be stable if the total number of electrons in the bonding molecular orbitals is greater than the number of electrons in the antibonding molecular orbitals. On the other hand, the molecule will be unstable if the number of electrons in the antibonding orbitals is more than the number of electrons in the bonding molecular orbitals.

(ii) **Bond order** It is the number of covalent bonds in the molecule and is equal to half the difference between the number of electrons in the bonding molecular orbitals and the number of electrons in the antibonding molecular orbitals. The stability of a molecule is determined by its bond order.

$$\text{Bond order} = \frac{\text{Number of electrons in bonding MOs} - \text{Number of electrons in antibonding MOs}}{2}$$

Single, double and triple bonds have the bond order of one, two and three, respectively. In some cases, bond order may be fractional.

Bond order conveys the following information

(a) A negative or zero bond order corresponds to an unstable molecule, whereas positive bond order indicates a stable molecule.

(b) Bond dissociation energy is directly proportional to the bond order. Higher bond dissociation energy means a stable molecule. For example, nitrogen and oxygen molecules with the bond orders of 3 and 2, respectively, have the bond dissociation energies of 945 and 495 kJ mol^{-1}, respectively.

(c) Bond order is inversely proportional to bond length. For example,

Molecule	Bond order	Bond length
O_2	2	121 pm
N_2	3	110 pm

(iii) **Magnetic Character** A molecule is diamagnetic if all the electrons are paired and paramagnetic if it has unpaired electrons.

Bonding in Some Homonuclear Diatomic Molecules

Let us discuss bonding in some homonuclear diatomic molecules of the elements of first and second rows of the periodic table.

1. **Hydrogen molecule (H_2)** Hydrogen molecule is formed by the combination of $1s^1$ atomic orbitals of two hydrogen atoms. The hydrogen molecule thus formed has two electrons that are accommodated in the bonding molecular orbital and the antibonding molecular orbital remains vacant. The molecular orbital energy level diagram for H_2 molecule is shown in Figure 10.44.

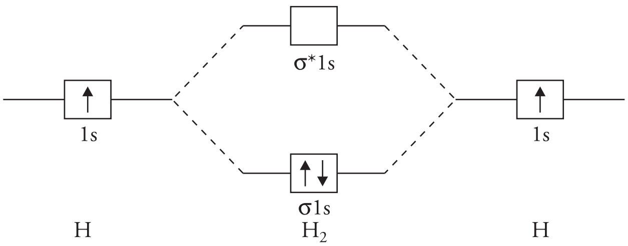


Figure 10.44 *Molecular orbital energy level diagram for H_2 molecule*

The electronic configuration of H_2 molecule is represented as follows

$H_2 : (\sigma 1s)^2$

The bond order in H_2 is

$$\text{Bond order} = \frac{N_b - N_a}{2} = \frac{2-0}{2} = 1$$

Bond order one means that there is a single covalent bond between the two hydrogen atoms. The bond length is found to be 74 pm and the bond dissociation energy is 485 kJ mol^{-1}. Absence of unpaired electron makes the molecule diamagnetic.

2. **Hydrogen molecule ion (H_2^+)** Hydrogen molecule ion is formed by the combination of a hydrogen atom with one electron and a hydrogen ion without an electron. The molecular orbital of H_2^+ ion contains one electron that occupies the bonding molecular orbital. Presence of an unpaired electron makes the molecule paramagnetic. The molecular orbital diagram is given in Figure 10.45.

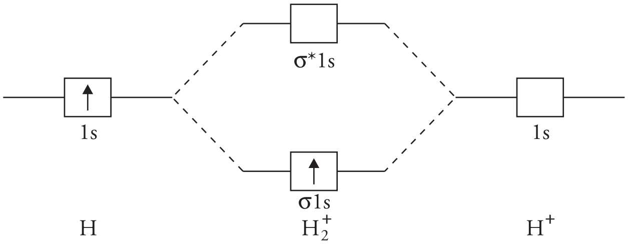


Figure 10.45 *Molecular orbital energy level diagram for H_2^+ ion*

The electronic configuration of the molecule is $H_2^+ : (\sigma 1s)^1$

$$\text{Bond order} = \frac{N_b - N_a}{2} = \frac{1-0}{2} = \frac{1}{2}$$

As the bond order has a positive value, the molecule is stable. The bond length of $\mathbf{H_2^+}$ ion is larger than that of H_2 molecule (bond length $\mathbf{H_2^+}$ =104 pm, H_2 = 74 pm), whereas its bond dissociation energy is less than that of H_2 molecule (bond dissociation energy $\mathbf{H_2^+}$ = 269 kJ mol^{-1}, H_2 = 485 kJ mol^{-1}). This clearly supports the fact that bond in $\mathbf{H_2^+}$ ion is weaker than that in H_2 molecule.

3. Hydrogen molecule negative ion H_2^-

A hydrogen atom with one electron in its valence shell and a hydrogen ion with two electrons combine to form H_2^- ion. Thus, H_2^- ion has three electrons and its electronic configuration will be $(\sigma 1s)^2$ $(\sigma^* 1s)^1$ and

$$\text{Bond order} = \frac{1}{2}(2-1) = \frac{1}{2}$$

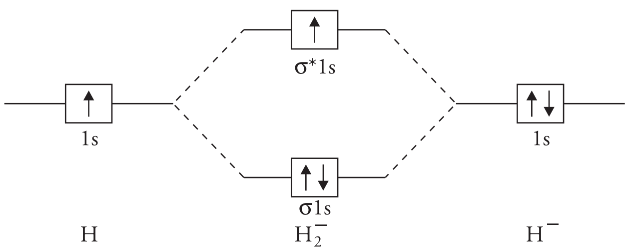


Figure 10.46 *Molecular orbital energy level diagram for H_2^- ion*

4. Hypothetical helium molecule (He_2) Helium atom has the electronic configuration $1s^2$; therefore, He_2 molecule has four electrons in it. These electrons occupy σ 1s and σ*1s MOs as shown in Figure 10.47. The electronic configuration for helium molecule may be written as

$$He_2 : (\sigma\, 1s)^2\, (\sigma^*\, 1s)^2$$

$$\text{Bond order} = \frac{N_b - N_a}{2} = \frac{2-2}{2} = 0$$

The **bond order is zero** that indicates that there is no net bonding and He_2 molecule will not be formed and the same has been proved experimentally.

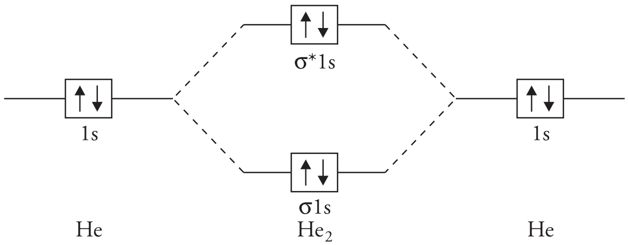


Figure 10.47 *Molecular orbital energy level diagram for He_2 molecule (hypothetical)*

5. **Helium molecule ion (He_2^+)** This molecule is formed by the combination of a helium atom containing two electrons and He^+ ion with one electron. The He_2^+ ion thus has three electrons that occupy the $(\sigma 1s)^2$ and $(\sigma^* 1s)^1$ orbitals. The molecular orbital electronic configuration of the molecule is

   $He_2^+ : (\sigma 1s)^2 (\sigma^* 1s)^1$

   $$\text{Bond order} = \frac{2-1}{2} = \frac{1}{2}$$

   Positive value of bond order indicates that He_2^+ is stable and its bond dissociation energy is found to be 242 kJ mol^{-1}. The presence of an unpaired electron makes the molecule paramagnetic.

Lithium molecule The valence shell configuration of lithium atom is $1s^2 2s^1$. Thus, lithium molecule Li_2 has six electrons. These are arranged as $(\sigma 1s)^2 (\sigma^* 1s)^2 (\sigma 2s)^2$. The inner shells do not contribute to bonding; hence, the electronic configuration can be written as $KK\,\sigma 2s^2$ where KK denotes that K shell of the two atoms is complete and does not contribute to bonding.

$$\text{Bond order} = \frac{1}{2}(4-2) = 1$$

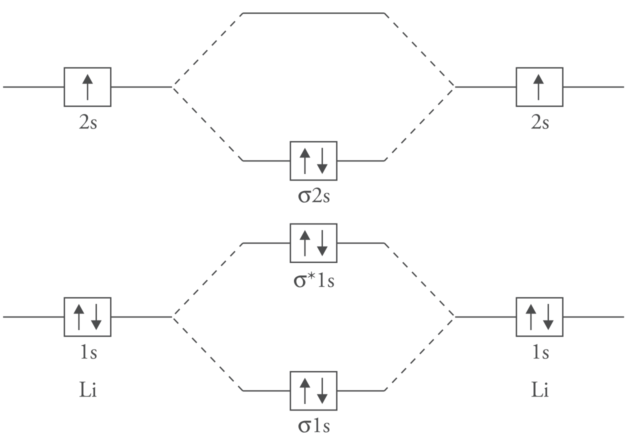


Figure 10.48 *Energy level diagram of Li_2*

Beryllium molecule Its electronic configuration is $1s^2 2s^2$. Be_2 has eight electrons; hence, its molecular orbital configuration will be $(\sigma 1s)^2(\sigma^* 1s)^2(\sigma 2s)^2(\sigma^* 2s)^2$ or it can be written as $KK(\sigma 2s)^2(\sigma^* 2s)^2$

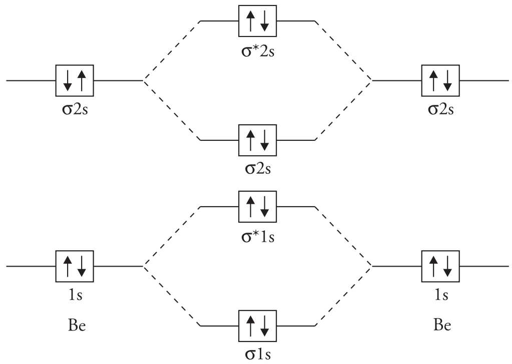


Figure 10.49 *Molecular orbital energy level diagram for Be_2*

$$\text{Bond order} = \frac{4-4}{2} = 0$$

Bond order is zero, hence Be_2 molecule is not formed.

Boron molecule (B_2) The electronic configuration of boron is $1s^2\ 2s^2\ 2p^1$. B_2 has 10 electrons. Its molecular orbital configuration will be $KK(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p_x)^1 = (\pi 2p_y)^1$.

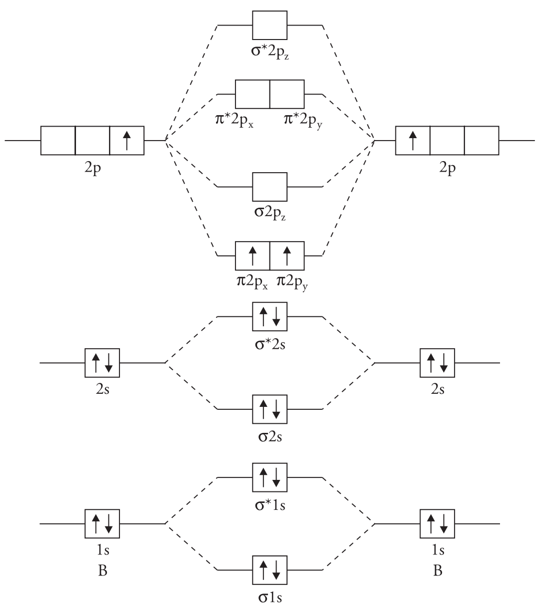


Figure 10.50 *Molecular orbital energy level diagram for B_2*

$$\text{Bond order} = \frac{1}{2}(6-4) = 1$$

As there are two unpaired electrons the molecule is paramagnetic.

Carbon molecule (C_2) A carbon atom has six electrons and its electronic configuration is $1s^2 2s^2 2p^2$. C_2 molecule will have 12 electrons. Its electronic configuration will be $KK\,(\sigma 2s)^2\,(\sigma^* 2s)^2\,(\pi 2p_x)^2 = (\pi 2p_y)^2$.

$$\text{Bond order} = \frac{1}{2}(8-4) = 2$$

Although it appears that C_2 is stable, carbon exists as macromolecule in graphite and diamond. In diamond and graphite, each carbon forms four bonds in preference to C_2.

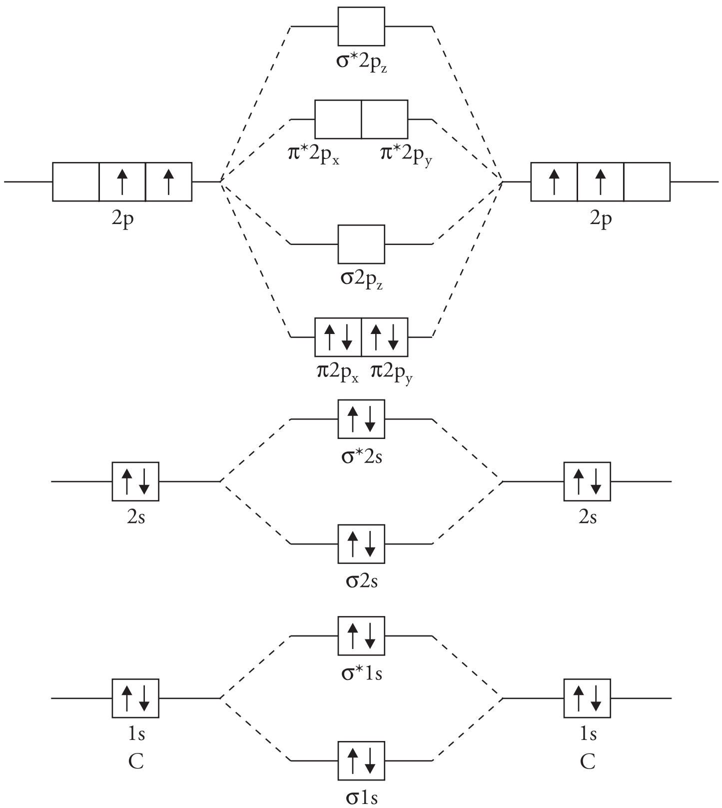


Figure 10.51 *Molecular orbital energy level diagram for C_2*

Table 10.2 summarises the molecular orbital configuration, bond order and magnetic character of Li_2, Be_2, B_2 and C_2.

Table 10.2 Molecular orbital configurations of Li_2, Be_2, B_2 and C_2

Molecule	MO configuration	Bond order	Magnetic character
Li_2	KK $(\sigma 2s)^2$	1	Diamagnetic
Be_2	KK $(\sigma 2s)^2$ $(\sigma^* 2s)^2$	0	Molecule does not exist
B_2	KK $(\sigma 2s)^2(\sigma^* 2s)^2(\pi 2p_x)^1 = (\pi 2p_y)^1$	1	Paramagnetic
C_2	KK $(\sigma 2s)^2\ (\sigma^* 2s)^2(\pi 2p_x)^2 = (\pi 2p_y)^2$	2	Diamagnetic

6. **Nitrogen molecule (N_2)** Each of the two nitrogen atoms with an electronic configuration of $1s^2\ 2s^2\ 2p_x^1\ 2p_y^1\ 2p_z^1$ contributes five valence electrons to the N_2 molecule. The MO diagram is shown in Figure 10.52.

The molecular orbital configuration of the molecule is

$$N_2 : KK\,(\sigma 2s)^2\,(\sigma^* 2s)^2\,(\pi 2p_x)^2 = (\pi 2p_y)^2\,(\sigma 2p_z)^2$$

$$\text{Bond order} = \frac{8-2}{2} = 3$$

Thus, there are three bonds in the nitrogen molecule, one σ and two π bonds. This is in accordance with very high bond dissociation energy of 945 kJ mol^{-1} and small bond length of 110 pm. The molecule is diamagnetic because it has no unpaired electrons.

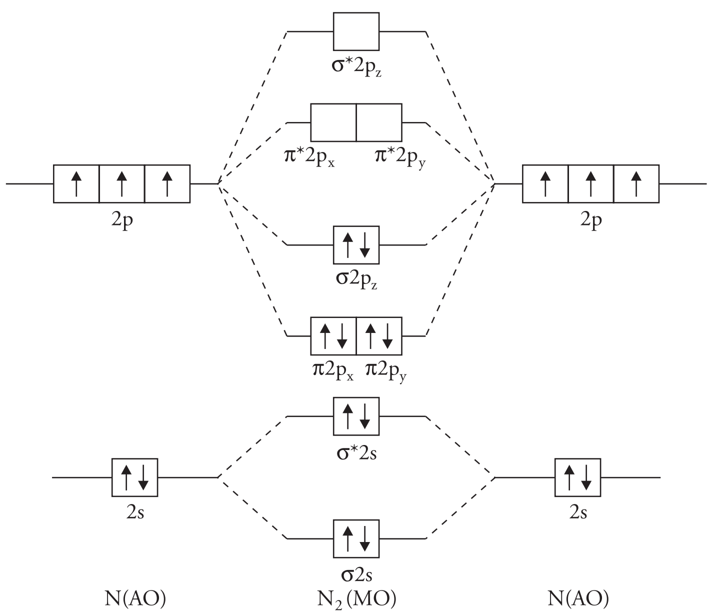


Figure 10.52 *Molecular orbital energy level diagram for N_2 molecule*

N_2^+ molecule and its comparison with N_2

N_2 molecule loses one electron to form the N_2^+ ion. This electron is lost from $\sigma 2p_z$ MO and hence the electronic configuration of $\mathbf{N_2^+}$ ion will be

$$N_2^+ : KK\,(\sigma 2s)^2\,(\sigma^* 2s)^2\,(\pi 2p_x)^2 = (\pi 2p_y)^2\,(\sigma 2p_z)^1$$

$$\text{Bond order} = \frac{7-2}{2} = 2\frac{1}{2}$$

Bond order of N_2^+ ion is $2\frac{1}{2}$ and the bond order of N_2 molecule is 3; hence, the N_2^+ bond is longer and weaker than that of N_2 molecule.

7. **Oxygen molecule (O_2)**

Oxygen atom with the electronic configuration of $1s^2\ 2s^2\ 2p_x^2 2p_y^1 2p_z^1$ has six electrons in its valence shell. The molecular orbitals formed by the overlap of inner 1s atomic orbitals are denoted by KK and it signifies $(\sigma 1s)^2\ (\sigma^* 1s)^2$. The remaining 12 electrons are filled in molecular orbitals as shown in Figure 10.53.

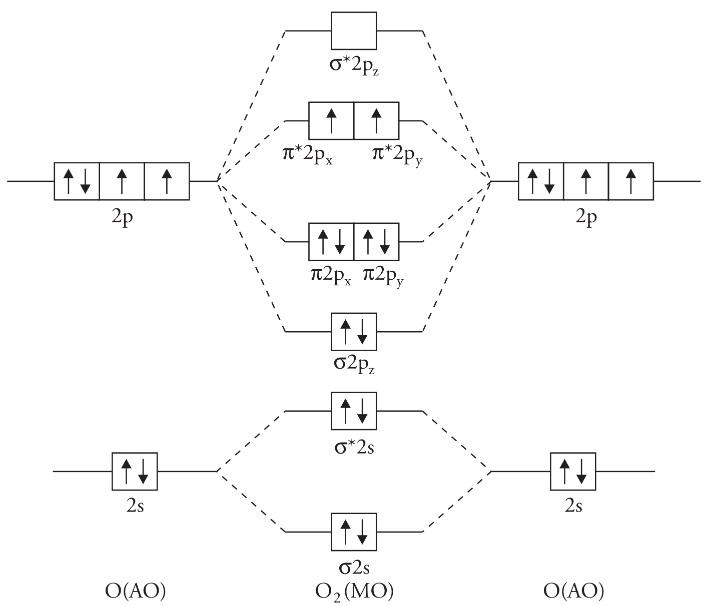


Figure 10.53 *Molecular orbital energy level diagram for O_2 molecule*

The electronic configuration of the molecule is

$$O_2 : KK\,(\sigma 2s)^2\,(\sigma^* 2s)^2\,(\sigma 2p_z)^2\,(\pi 2p_x)^2 = (\pi 2p_y)^2\,(\pi^* 2p_x)^1 = (\pi^* 2p_y)^1$$

$$\text{Bond order} = \frac{8-4}{2} = 2$$

The bond order of two denotes that the oxygen molecule has a double bond (one σ and other π). It has two unpaired electrons in π^*2p_x and π^*2p_y orbitals, which makes the molecule paramagnetic. All the above facts have been proved experimentally. The bond dissociation energy of O_2 molecule is 495 kJ mol^{-1} and the bond length is 121 pm.

Comparison of O_2, O_2^+ and O_2^- species

8. **Oxygen molecule ion (O_2^+)** When one electron is removed from O_2 molecule, O_2^+ ion is formed. The electron will be lost from antibonding MO, that is, π^*2p_x or π^*2p_y

$$O_2 \longrightarrow O_2^+ + e^-$$

The electronic configuration of O_2^+ is

$$O_2^+ : KK(\sigma 2s)^2(\sigma^*2s)^2(\sigma 2p_z)^2(\pi 2p_x)^2 = (\pi 2p_y)^2(\pi^*2p_x)^1$$

$$\text{Bond order} = \frac{8-3}{2} = \frac{5}{2} = 2\frac{1}{2}$$

The bond order of O_2^+ ion is greater than the bond order of O_2 molecule; therefore, the bond strength of O_2^+ will be more than that of O_2 molecule and its bond length will be less than the bond length of O_2 molecule.

Superoxide ion (O_2^-) Addition of one electron to O_2 forms O_2^-. This electron is added to either of the antibonding MO (s) $(\pi^*2p_x \text{ or } \pi^*2p_y)$

$$O_2 + e^- \longrightarrow O_2^-$$

The electronic configuration of O_2^- is

$$O_2^- : KK(\sigma 2s)^2(\sigma^*2s)^2(\sigma 2p_z)^2(\pi 2p_x)^2 = (\pi 2p_y)^2(\pi^*2p_x)^2 = (\pi^*2p_y)^1$$

$$\text{Bond order} = \frac{8-5}{2} = \frac{3}{2} = 1\frac{1}{2}$$

The bond order of O_2^- ion is less than the bond order of O_2 molecule; therefore, its bond strength is less than O_2 molecule and bond length is larger.

Peroxide ion (O_2^{2-}) When two electrons are added to the antibonding orbitals of O_2 molecule, a peroxide ion O_2^{2-} is obtained.

$$O_2 + 2e^- \longrightarrow O_2^{2-}$$

The molecular orbital configuration of O_2^{2-} ion is

$$O_2^{2-} : KK(\sigma 2s)^2(\sigma^*2s)^2(\sigma 2p_z)^2(\pi 2p_x)^2 = (\pi 2p_y)^2(\pi^*2p_x)^2 = (\pi^*2p_y)^2$$

$$\text{Bond order} = \frac{8-6}{2} = 1$$

The bond order of O_2^{2-} ion is less than the bond order of O_2 molecule; therefore, its bond strength is less than O_2 molecule and bond length is larger.

Bond dissociation energies and bond lengths for O_2^+, O_2 and O_2^- are in Table 10.3.

Table 10.3 Bond dissociation energies and bond lengths for O_2^+, O_2 and O_2^-

Species	**Bond order**	**Bond dissociation energy (kJ mol^{-1})**	**Bond length**
O_2^+	$2\frac{1}{2}$	625	112 pm
O_2	2	495	121 pm
O_2^-	$1\frac{1}{2}$	395	130 pm

The species O_2, O_2^+, O_2^- and O_2^{2-} can be arranged as

Bond dissociation energy $O_2^+ > O_2 > O_2^- > O_2^{2-}$

Bond length $O_2^{2-} > O_2^- > O_2 > O_2^+$

9. **Fluorine molecule (F_2)** Fluorine atom with an electronic configuration of $1s^2\ 2s^2\ 2p^5$ has seven electrons in its valence shell. The molecular orbital electronic configuration of F_2 molecule is

$$F_2 = KK(\sigma 2s)^2(\sigma^* 2s)^2(\sigma 2p_z)^2(\pi 2p_x)^2 = (\pi 2p_y)^2(\pi^* 2p_x)^2 = (\pi^* 2p_y)^2$$

$$\text{Bond order} = \frac{8-6}{2} = 1$$

Thus, there is one σ bond in the molecule. The molecular orbital diagram shows that all the electrons are paired, making the molecule diamagnetic. Its diamagnetic behavior has been proved experimentally. The bond dissociation energy is found to be 155 kJ mol^{-1} and bond length is 142 pm.

10. **Hypothetical neon molecule** Neon atom has the electronic configuration $1s^2\ 2s^2\ 2p^6$. The molecular orbital electronic configuration of neon molecule will be

$$Ne_2 : KK(\sigma 2s)^2(\sigma^* 2s)^2(\sigma 2p_z)^2(\pi 2p_x)^2 = (\pi 2p_y)^2(\pi^* 2p_x)^2 = (\pi^* 2p_y)^2(\sigma^* 2p_z)^2$$

$$\text{Bond order} = \frac{8-8}{2} = 0$$

Thus, Ne_2 molecule is non-existent.

Heteronuclear diatomic molecules

The electronegativities of the two combining atoms in heteronuclear diatomic molecules are different. When such molecules form molecular orbitals, the bonding molecular orbital lies closer to the atom with higher electronegativity and the antibonding molecular orbital lies closer to the less electronegative atom.

In homonuclear molecules, the energy of the combining orbitals is equal, hence they interact effectively; however, in heteronuclear molecules the energy of the combining atomic orbitals is different hence effective overlap is not possible. Let us discuss the electronic structure and molecular orbital diagrams for some heterogeneous diatomic molecules.

1. **Hydrogen Fluoride molecule, HF** The electronic configuration of hydrogen and fluorine atoms are

$H = 1s^1$

$F = 1s^2 2s^2 2p^5$

HF molecule is formed by the linear combination of the H(1s) atomic orbital and one of the F atomic orbitals. The $1s^2$ and $2s^2$ orbitals of fluorine do not take part in bonding because their energy is very low. Because of symmetry considerations, out of the three 2p orbitals, only $2p_z$ orbital is able to combine with an s orbital. Therefore, during the formation of HF molecule, the bonding occurs due to the overlap of H (1s) electrons and $2p_z$ electrons of F, and the other electrons of fluorine remain in atomic orbitals.

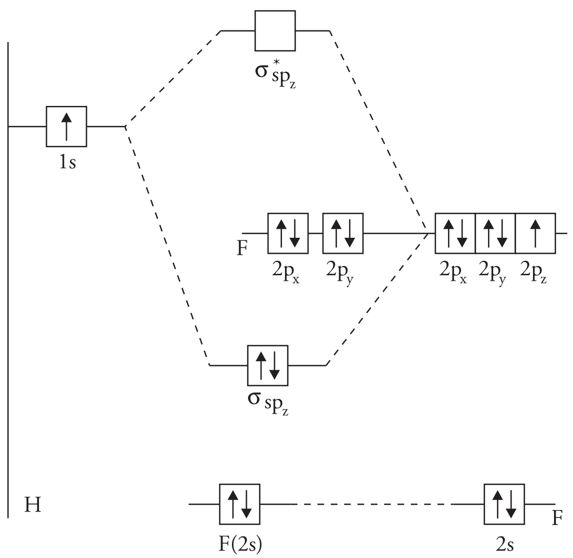


Figure 10.54 *Molecular orbital energy level diagram for HF*

The structure of HF molecule, therefore, may be written as

$$H(1s^1) + F(1s^2\,2s^2\,2p^5) \rightarrow HF[K(2s)^2(\sigma sp_z)^2(2p_x)^2(2p_y)^2]$$

It is important to note that only one bonding molecular orbital σ_{2spz} is formed in HF molecule. This bonding MO contains two electrons. The remaining electrons remain in their atomic orbitals $2p_x$ and $2p_y$. The configuration of HF molecule closely resembles that of the neon atom ($1s^22s^22p^6$); however, as HF molecule is heteronuclear, the electrons are no longer shared equally between the two nuclei. The F($2p_z$) atomic orbital is contributing more than the H (1s) orbital to the molecular orbitals. As seen in Figure 10.54 the molecular orbital has an energy that lies closer to that of $2p_z$ orbital of fluorine than to that of 1s orbital of hydrogen.

Similarly, in HCl, HBr and HI the electrons are shared unequally between the nuclei of their atoms. The chlorine, bromine and iodine atoms contribute, respectively, $3p_z$, $4p_z$ and $5p_z$ atomic orbitals in formation of these molecules.

The molecular orbital diagram for HCl will be similar to HF.

2. **Nitric oxide molecule (NO)** The electronic configuration of N and O atoms are

$$N = 1s^22s^22p_x^{\ 1}2p_y^{\ 1}2p_z^{\ 1}$$

$$O = 1s^22s^22p_x^{\ 2}2p_y^{\ 1}2p_z^{\ 1}$$

The outer shell of nitrogen and oxygen have five and six electrons, respectively; thus, NO molecule has a total of 11 electrons. MO diagram for NO molecule is shown in Figure 10.55. The bonding MOs are closer to oxygen, as its electronegativity value is higher than nitrogen and antibonding MOs are closer to N atom.

The molecular orbital electronic configuration of NO molecule is

$$NO = KK(\sigma 2s)^2(\sigma^* 2s)^2(\sigma 2p_z)^2(\pi 2p_x)^2 = (\pi 2p_y)^2(\pi^* 2p_x)^1$$

$$\text{Bond order } = \frac{8-3}{2} = 2\frac{1}{2}$$

The presence of an unpaired electron makes the NO molecule paramagnetic. The molecular orbital diagram of NO is similar to that of diatomic molecules such as O_2 or F_2, except that the energy level of N and O are not the same.

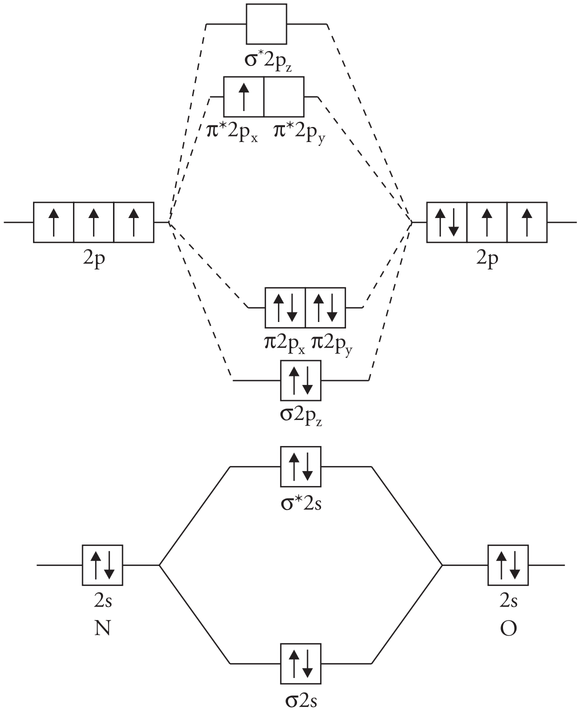


Figure 10.55 *Molecular orbital diagram for NO molecule*

3. **Carbon monoxide molecule CO** The electronic configuration of carbon and oxygen atoms are

$C = 1s^2 2s^2 2p^2$ and $O = 1s^2 2s^2 2p^4$

The outer shell of carbon and oxygen have four and six electrons, respectively; thus the CO molecule has 10 electrons that occupy its molecular orbitals. The MO diagram for CO molecule may be drawn similar to NO molecule as shown in Figure 10.56. Then the molecular orbital configuration may be written as

$$CO = KK(\sigma 2s)^2 (\sigma * 2s)^2 (\sigma 2p_z)^2 (\pi 2p_x)^2 = (\pi 2p_y)^2$$

$$\text{Bond order } = \frac{8-2}{2} = 3$$

Thus, CO molecule has a triple bond (one σ and two π bonds). The molecule is diamagnetic, as there is no unpaired electron in it. Figure 10.56 gives the molecular orbital diagram of CO

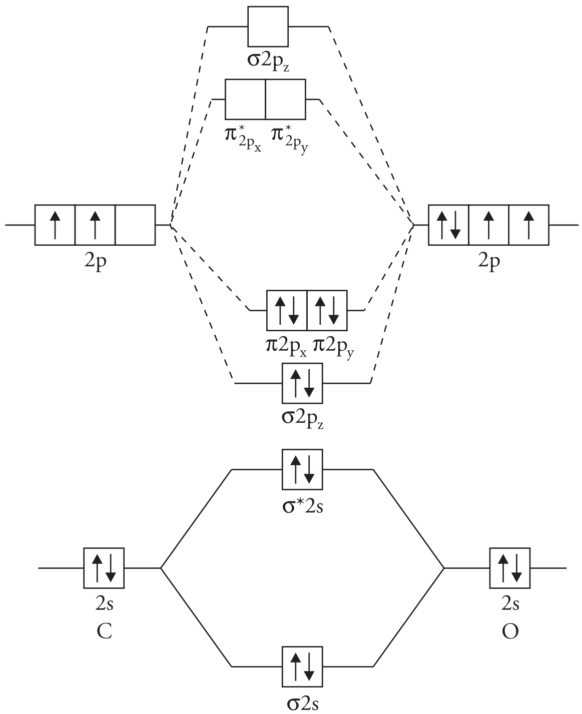


Figure 10.56 *Molecular orbital diagram for CO molecule*

10.13 Bonding in Metals – Metallic Bond

Metals have certain characteristic physical properties

1. They are good conductors of heat and electricity. The electrical conduction in metals is because of free electrons. It may be recalled that ionic crystals such as NaCl also conduct electricity, but in molten or fused state. In the solid state, ionic compounds conduct electricity to a very small extent because of the presence of defects in their crystals. The electrical conduction in metals is very different from the conduction of electricity in ionic solids.
2. Metals are lustrous. They have a shiny appearance.
3. Metals are malleable. This shows that they do not offer much resistance to deformation of the structure but that large cohesive forces hold them together.
4. The melting and boiling points of metals are high.
5. They are crystalline with high coordination number of 12 or 14.

All the above-mentioned properties of metals cannot be explained on the basis of normal ionic or covalent bonding.